SDCDNet: A Semi-Dual Change Detection Network Framework With Super-Weak Label for Remote Sensing Image

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***Abstract*— Most current change detection methods require a large amount of labeled data to train huge parameters. To break this limitation, this article proposes a novel semi-supervised learning (SSL) framework for remote sensing change detection, named a semi-dual change detection network (SDCDNet). The SDCDNet consists of a dual shared network and dual branching networks. The dual shared network is designed to exploit the full potential of the data, and the dual branching network is proposed to differentiate the kinds of annotated data and eliminate the disturbance between different types of data. In addition, the adaptive weighting module (AWM) enhances the features of weak branching, and the mask constraint module (MCM) is proposed to increase the ability of the network to extract foreground features. To solve the complex problem of data labeling, a patch-based weak label construction method is proposed to build super-weak labels. Experiments show that the proposed SDCDNet achieves excellent results on two remote sensing image change detection datasets.**

***Index Terms*— Change detection, dual branch network, remote sensing image, semi-supervised learning (SSL).**

I. Introduction

CHANGE detection has now become an important research direction in the field of remote sensing and has been widely applied in many fields [1], [2], [3], [4], [5], [6], [7], [8]. Especially in the fields of building damage detection [9], [10], environmental monitoring [11], [12], [13], disaster monitoring [14], [15], urban change analysis [16], [17], [18], [19] and land cover surveillance [20], [21], [22], [23], change detection technology performs a great role.

With the advances in remote sensing observation technology, various remote sensing images are easily accessible. The remote sensing data provided by different platforms can be used for analysis and research, providing strong support for our decision-making and management [24]. By comparing data from the same location at different times, we can have information about the changes in a specific area. And as the efficiency of change detection technology improves, the change detection technology will be of greater value [25].

The task of change detection is analyzing multitemporal image pairs and finally assigning a binary label to each pixel [26]. The allocated label "0" means that the surface corresponding to the pixel has not changed semantically, and conversely, the label "1" means that the surface corresponding to the pixel has changed semantically between T0 and T1. And it is very challenging to detect changes in very high-resolution (VHR) images due to seasonal variations, and imaging conditions [27].

Change detection techniques for multitemporal VHR remote sensing images are mainly classified into traditional methods and deep learning-based methods [28]. Many traditional change detection techniques, such as image algebra and transformations, are relatively limited in practical applications and are susceptible to the effects of the sensors themselves, atmospheric and seasonal variations, and solar altitude, which degrade the performance of change detection. Most traditional methods rely on hand-designed feature descriptors [20], [29], [30], [31], [32], [33] with limited ability to describe complex and high-level semantic change information, resulting in low performance under messy land cover. In recent years, various deep learning techniques have been widely used and numerous deep learning-based algorithms for remote sensing change detection have been proposed. Powerful data pattern modeling capability is the most important feature of deep learning-based methods [34], [35], [36]. Deep learning-based change detection methods can automatically mine abstract, complex, and nonlinear feature representations between input image pairs and achieve high change detection performance, and

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they can be subdivided into supervised, semi-supervised, and unsupervised methods.

Existing deep learning network models generally have more parameters, so the effectiveness of change detection methods depends heavily on the quality and quantity of labeled training data [37], especially for fully supervised methods. When we do not have a large amount of training data, the detection ability of change detection models is greatly reduced [38]. The real problem is that accurate annotation of remotely sensed images is a very expensive task [39]. On the one hand, annotation requires experts with certain expertise; on the other hand, there may be additional costs for parts that are not easy to annotate.

With the evolvement of deep learning, many scholars have realized that relying on a large amount of finely labeled data is unsustainable. Therefore, lots of academics have focused their research on the problem of how to learn using a small number of annotations, such as semi-supervised learning (SSL), weakly-supervised learning, self-supervised learning, and unsupervised learning. And based on these learning approaches, many novel change detection methods have been proposed. SSL is a learning approach that combines the advantages of supervised and unsupervised learning, and it uses a large amount of unlabeled data with as little labeled data as possible while requiring to achieve performance no less than that of fully supervised methods, so SSL is now receiving more and more attention [40].

For the change detection study, there is currently no unified framework for extending fully supervised methods to semi-supervised data conditions, and implementing such approaches can be challenging due to their complexity. To address the lack of a large amount of high-precision annotated data in practical applications, a semi-supervised dual change detection network (SDCDNet) framework with an adaptive weighting module (AWM) and a mask constraint module (MCM) is designed. The proposed framework can use both very limited pixel-level labeled data and a large amount of patch-level labeled data in training. The AWM in the framework adaptively weights the features at different semantic levels of the feature extraction network, and the MCM constrains the features of the weak branches of the input to give more attention to the change regions. Our experimental results show that the framework can achieve higher accuracy change detection with less fine annotation, and the proposed two modules can effectively direct the network's attention to useful change information, and the ablation experiments are described in Section IV.

The contributions of this article are as follows.

1) A novel SSL framework is proposed for the change detection task. Our method perfectly adapts to most fully supervised methods and achieves similar scores as fully supervised methods with very limited normal labeled data.
2) To make full use of unlabeled data, a super-weak label construction method combined with patches is proposed, the size of which can be adjusted by ourselves, and the super-weak labels are fed into the proposed semi-supervised network together with the normal labels for training.
3) To obtain better foreground features for the weakly labeled branches of the framework, an AWM and MCM are proposed to extract features that are more useful for the change detection task.

This article is organized as follows. In Section II, some related works are presented. Section III describes our proposed method in detail. Section IV gives the setup and results of several test experiments and ablation experiments to corroborate the effectiveness of our proposed method. Section V concludes this article.

II. Related Work

In the works of literature, many methods have been published in remote sensing images for the subject of change detection. In this section, we introduce the applications of fully-supervised and semi-supervised methods in the field of change detection, respectively.

A. Fully Supervised Method

For fully supervised methods, some of them use fully convolutional networks (FCNs) to generate a high-resolution change map from the bitemporal inputs, which is widely applicated in remote sensing image understanding. Among them, a part of the approaches concatenate two images and feed it into change detection networks [41], which is called early fusion. On the other hand, some other methods use two-branch networks to extract deep features and then fusion and generate the change map.

In [42], the FC-Siam-Diff method based on the U-Net framework for change detection is proposed and it first inputs the image pairs into the siamese network and uses the feature differences of multilevel features extracted from UNet to generate the final change map, realizing the change detection of remote sensing image pairs. In [43], a spatial–temporal attention neural network was proposed to implement change detection, and the spatial-temporal attention mechanism calculates the attention between two pixels at different times and spaces to generate more discriminative features. In [44], a difference-enhancement dense-attention network was used to integrate the low-level and high-level features.

Liu et al. [45] use a multitask learning approach for the training of the change detection network to solve the problems of incomplete detection result regions or irregular region boundaries. The proposed dual-task constrained deep siamese convolutional network (DTCDSCN) comprises a change detection network and two semantic segmentation networks.

Fang et al. [46] proposed a densely connected siamese network with channel attention for change detection. Zhang et al. [47] proposed a deep supervised dual-branch image fusion FCN for remote sensing image change detection. The method extracts dual-temporal image deep features and uses the deep-supervised differential recognition network to guide the training. Chen et al. [48] combine siamese architecture with recurrent neural network (RNN) to realize higher performance for multisource data. To obtain more discriminant feature representations, in [49], a siamese FCN with a dual attention mechanism was proposed, and in [50], a hierarchical

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dynamic fusion strategy was proposed to provide more useful information. In [51], the self-attention mechanism was used to model the semantic relations between any two pixels in bitemporal space-time.

With the potential of Transformers explored in remote sensing, some literature about change detection based on transformers was proposed by researchers. In [52], a change detection method based on Transformers was proposed. This method uses convolutional neural network (CNN) to extract deep features and feed them into bitemporal image transformer (BIT). The authors implement BIT to learn and relate the global information of high-level semantic concepts so that the original bitemporal features are enhanced. Then the element-wise absolute of the subtraction of the bitemporal feature maps is generated. Finally, prediction head based on convolution are leveraged to predict the change map.

B. Semi-Supervised Method

The remote sensing image labeling is difficult and expensive, so the approach that achieve specific performance and relies on less data is more valuable.

Some of the semi-supervised methods are based on GANs. In [53], a semi-supervised change detection network based on FCN and GAN is proposed. The authors use UNet++ with attention as the generator of GAN and use two discriminators to reward the output feature distribution consistency and suppress the uncertainty region in the change map of unlabeled data, respectively. By continuously learning from both labeled and unlabeled data, the generator can generate expected change maps. In [54], a semi-supervised semantic segmentation network was proposed. It consists of a segmentor and a discriminator, the segmentor and the discriminator are trained in an adversarial way with a proposed self-training loss and feature-matching loss. The feature-matching loss stabilizes low-data training and the self-training loss balances the segmentor and discriminator. In [55], a self-supervised conditional generative adversarial network (GAN) is proposed, which is trained using the mutual supervision information of generators and discriminators, and the change detection of dual-temporal remote sensing images is realized using the discriminators in GAN as the classifiers for change detection after the training is completed. Wu et al. [56] proposed an end-to-end change detection framework, which contains three basic modules: segmenter, discriminator, and generator. By combining two or two of them with different optimization strategies, the change detection task can be achieved under different supervision conditions.

Due to the training difficulty and stability of GANs, The non-GAN-based or more stable semi-supervised method is still a research hotspot of scholars. In [25], a graph convolutional network (GCN) was used to achieve semi-supervised change detection. In [57] a semi-supervised method for change detection is proposed. It contains two stages: fully supervised training and unsupervised training. The fully supervised training phase is a normal image segmentation training process. In the unsupervised training stage, the input image pairs have no ground truth change mask. Then first apply random perturbations on the deep feature difference map. Therefore constrains the output change maps to be consistent under different perturbations. Ismail and Awad [58] use GCN to learn discriminative features in a semi-supervised framework with few annotated data. In [39], a label refinement method was proposed to generate high-resolution change map from low-resolution data labeling. Sun et al. [59] train a dual-branch nested UNet with graph attention in a low-labeled data situation, and then retrain the network by constraining the consistency of the detection result of distorted image and the pseudo label of unlabeled data. In [60], a dual-task network integrated urban building segmentation and change detection and then implemented semi-supervised change detection.

On account of the complexity of remote sensing images, annotation for remote sensing images usually requires professional expertise, which results in the acquisition of pixel-level annotation data required by fully supervised methods being very expensive. In contrast, many nonfully supervised change detection methods require only a portion of pixel-level labeled data for training. They have achieved similar results in change detection tasks. Still, existing semi-supervised change detection methods are usually unable to fully exploit the potential of all data [61], [62], resulting in often unsatisfactory change detection when the proportion of pixel-level labeled data is low. In summary, the existing methods suffer from the following problems.

1) The pixel-level labeling supervision information of fully supervised methods is not easy to obtain.
2) The framework of existing semi-supervised methods is generally complex and lacks simple and effective strategies.
3) The supervisory information is not thoroughly explored, and the information in the precious fine-labeled data is not well utilized.

In response to the challenges outlined above, we propose a straightforward and effective framework that combines pixel-level and patch-level annotation for change detection tasks using the original fully supervised approach. By strategically utilizing and leveraging supervised information, we progressively enhance the consistency of change feature distribution between labeled and unlabeled data during training, thereby enabling the network to learn crucial features for change detection and achieving high performance with minimal pixel-level labeled data.

III. Method

In this section, we present the general structure of our proposed framework in Section III-A, which utilizes strongly labeled training data with pixel-level labels and weakly labeled training data based on patch annotation. We also describe the construction of a super-weak label in Section III-B. To improve the foreground features of the weak branches, we propose the AWM and MCM techniques, which are explained in Sections III-C and III-D, respectively. In Section III-E, we introduce the loss function used in this framework, and in Section III-F, we detail the training process.

Our proposed framework has broad applicability and can be extended to incorporate additional weakly labeled training data by simply adding weak branches to the original fully

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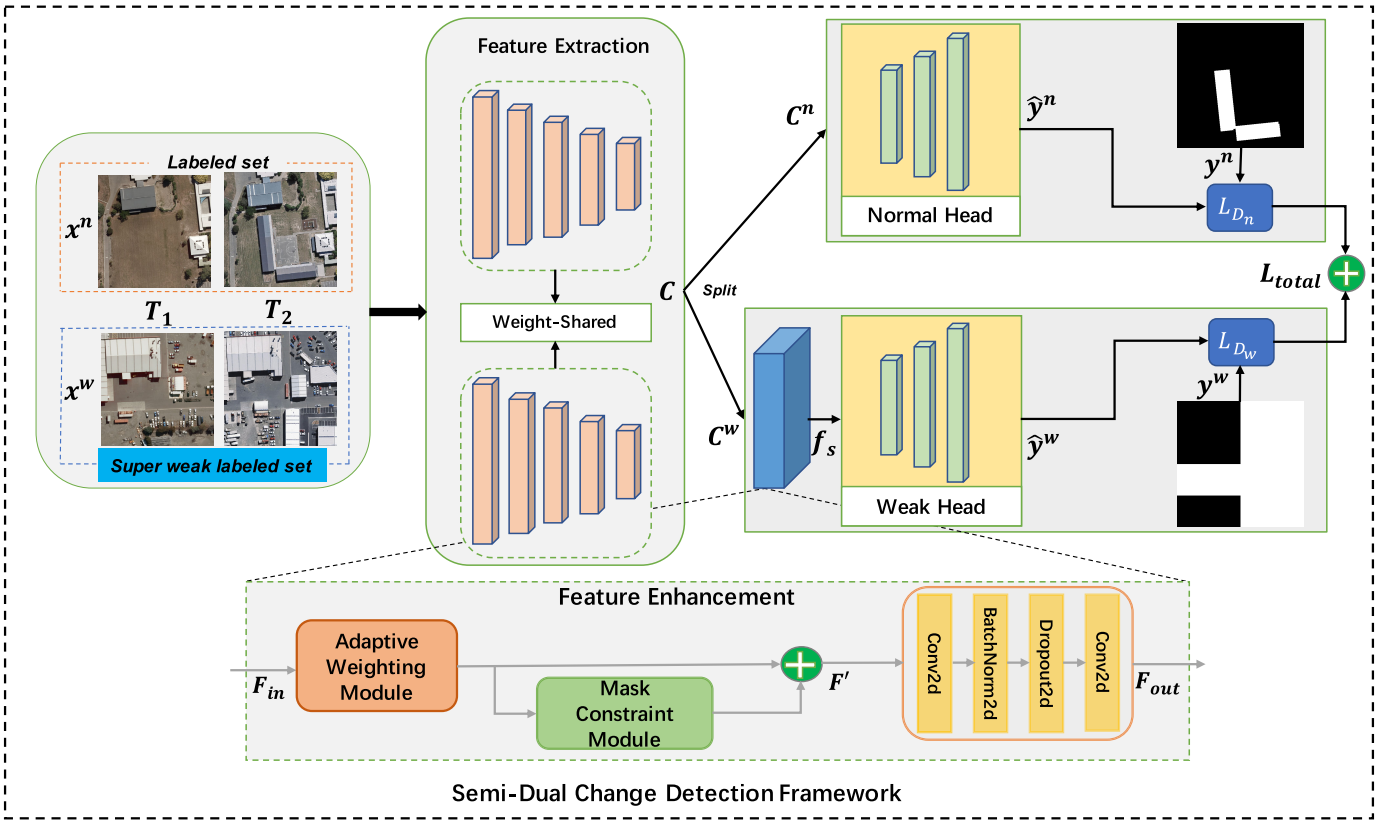


Fig. 1. Proposed framework for semi-supervised remote sensing image change detection, SDCDNet. The framework consists of a dual shared network and two branch networks. The proposed AWM and MCM can improve the ability of weak branches to process weak label information.

supervised network, without requiring any structural changes. Therefore, we can easily integrate our semi-supervised change detection framework with most existing Encoder-Decoder [63] paradigms for fully supervised change detection.

We aim to construct a semi-supervised remote sensing image change detection framework using a training dataset containing a strongly labeled training set and a weakly labeled training set. They are: 1) a minimal, fully supervised dataset in which all data have pixel-level labels corresponding to them, and 2) a dataset of the super-weak labels in which all data have patch-level labels corresponding to them.

A. Overall Structure

The general structure of our proposed method is shown in Fig. 1. It contains five parts: the construction of training data, the feature extraction module with double weight sharing, the AWM, the MCM, and the double branch module.

First, pixel-level strongly labeled images x^n and weakly labeled images x^w are sent simultaneously to a weight-sharing backbone network to generate features C, further C is split into C^n and C^w. The backbone network is mainly determined by the fully supervised change detection model, e.g., the backbone network in DSIFN [47] is VGG16 [64], the backbone network in SNUNet [46] is the encoder of UNet++ [65], etc. For Normal Head, C^n is fed directly. For Weak Head, since the corresponding weak annotation lacks a lot of detail information, the consistency constraint of the features C^w is weighted using the AWM to make a stronger multiscale consistency constraint in regions with low uncertainty, and then use the MCM to make Weak Head get better foreground features, thus, the generated f_s are fed to Weak Head. Finally, Normal Head and Weak Head generate the final predicted results $\hat{y}^n$ and $\hat{y}^w$, and then calculate the losses L_{D_n} and L_{D_w} with pixel-level annotations y^n and weak annotations y^w, respectively. $L_{D_n} + L_{D_w}$ is used for back-propagation training.

In the Weak Head, first, AWM is used to extract multiscale features adaptively. Then, to reduce the information loss and avoid the problem of gradient disappearance [66] in the training step, we add an identity branch in the module inspired by the residual connection in ResNet [61]. MCM is inserted into the branch to increase the ability of the network to extract foreground features. Finally, batch normalize [67] and dropout [68] are used to improve the network's convergence speed and generalization ability, respectively. As shown in Fig. 1, assuming $F_{\text{in}} = [F_1, F_2, \ldots, F_c] \in \mathbb{R}^{C\times H\times W}$, where C denotes the number of channels in the feature maps, H and W denote the height and width, respectively, and F' and F_{out} can be obtained from the following equations:

$$F' = \text{AWM}(F_{\text{in}}) + \text{MCM}(\text{AWM}(F_{\text{in}})) \tag{1}$$

$$F_{\text{out}} = \text{Conv}(D(B(\text{Conv}(F')))) \tag{2}$$

where D is dropout [68], B is batch normalize [67].

B. Construction of Super-Weak Labels

Accurate annotation for remote sensing image change detection is a costly affair. In this section, we propose a Patch-based

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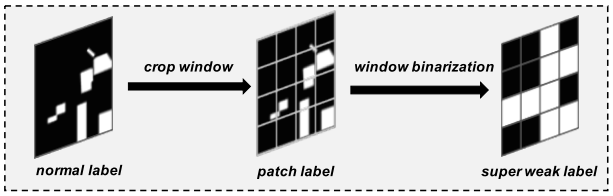


Fig. 2. Proposed construction method of super-weak labels based on patch, SWL. In the figure, we set the Patch Size to 128 pixels to divide the fine labels into four subsets of patches.

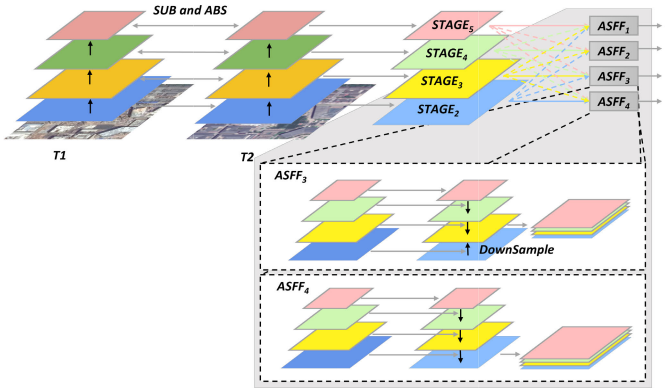


Fig. 3. Illustration of the proposed AWM.

super-weak labels generation method, SWL, which is no longer limited to the labeled population and reduces the labeling cost significantly.

First, we chunk the input labels by a predefined Patch Size. Second, for the kth patch P^k of size psize $\times$ psize, the finally generated weak patch label is defined as follows:

$$P^k_{\text{weak}} = 1 - \mathbb{I}\left(\sum_{i=1}^{\text{psize}}\sum_{j=1}^{\text{psize}} P^k_{ij} = 0\right) \tag{3}$$

where $\mathbb{I}(\cdot)$ is the indicator function with a value of 1 if $\cdot$ is true and 0 otherwise. P^k_{ij} represents the value of the (i, j) first pixel in the kth patches. Intuitively, when a patch contains pixel labels with changes, the whole patch takes the value of 1; when the whole patch contains no changes, each pixel in this patch takes the value of 0. In Fig. 2, the patch in the top-left corner has no pixels with changes, so its corresponding weak label will be given 0. In contrast, the remaining three patches contain change information, so their related weak labels will be assigned a value of 1. Finally, we get the inexactly labeled weak labels after the weak label construction process.

C. Adaptive Weighting Module

As shown in Fig. 3, the features of each stage obtained by feature extraction are fed into this module. First, for each stage's features, the remaining stages' features are convoluted and sampled to the same size as the features of the stage, and spliced with it, so that the feature information of different scales can be extracted from each stage. Then the same stage's features in different time sequences are passed through the adaptively spatial feature fusion (ASFF) module [69] to obtain the additional enhanced features of each final stage. For a backbone network with n stages, the specific enhanced feature calculation for each phase is expressed as follows:

$$\begin{aligned}
\text{weight}_1 &= \text{CS}(\text{Concat}(F_1, \text{DS}(F_2), \text{DS}(F_3), \text{DS}(F_4))) \\
\text{weight}_2 &= \text{CS}(\text{Concat}(\text{US}(F_1), F_2, \text{DS}(F_3), \text{DS}(F_4))) \\
\text{weight}_3 &= \text{CS}(\text{Concat}(\text{US}(F_1), \text{US}(F_2), F_3, \text{DS}(F_4))) \\
\text{weight}_4 &= \text{CS}(\text{Concat}(\text{US}(F_1), \text{US}(F_2), \text{US}(F_3), F_4)) \\
\text{ASFF}_i &= \text{weight}_i * F_i, \quad i = 1, 2, 3
\end{aligned} \tag{4}$$

where DS and US represent down-sampling and up-sampling operations, CS means that the Conv is performed first, followed by the SoftMax. The details can be found in [69]. The subsequent processing is done using the feature difference or feature splicing in change detection.

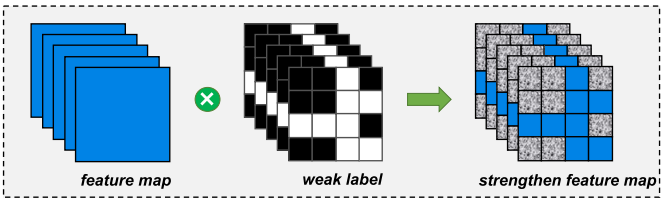


Fig. 4. Illustration of the proposed MCM.

D. Mask Constraint Module

In this module, the extracted change features are multiplied with the change foreground mask map provided by the weak label, invalidating the features in the unchanged region and making the network more concerned with feature extraction in the foreground of the transformed region, as shown in Fig. 4. The whole process can be expressed as

$$f \times P = f' \tag{5}$$

where f and f' represent feature map and strengthen feature map, respectively. P represents weak label, the white part is the change area, represented by 1, and the black part is represented by 0.

E. Loss Function

To train this framework, we mainly use the binary cross-entropy (BCE) loss [70]. L_{D_n} is used to denote the supervised loss, and L_{D_w} is used to indicate the weakly supervised loss, specifically for the samples in dataset k

$$\begin{aligned}
L_{D_n} = -\frac{1}{N^k}\sum_{m=1}^{N^k}\sum_{i,j}\Big(&y^{n(m)}_{i,j}\log\left(\hat{y}^{n(m)}_{i,j}\right) \\
&+\left(1 - y^{n(m)}_{i,j}\right)\log\left(1 - \hat{y}^{n(m)}_{i,j}\right)\Big)
\end{aligned} \tag{6}$$

$$\begin{aligned}
L_{D_w} = -\frac{1}{N^k}\sum_{m=1}^{N^k}\sum_{i,j}\Big(&y^{w(m)}_{i,j}\log\left(\hat{y}^{w(m)}_{i,j}\right) \\
&+\left(1 - y^{w(m)}_{i,j}\right)\log\left(1 - \hat{y}^{w(m)}_{i,j}\right)\Big)
\end{aligned} \tag{7}$$

where $\hat{y}^{n(m)}_{i,j}$ and $\hat{y}^{w(m)}_{i,j}$ represent the confidence maps predicted at location (i, j) obtained by SDCDNet of the sample m of

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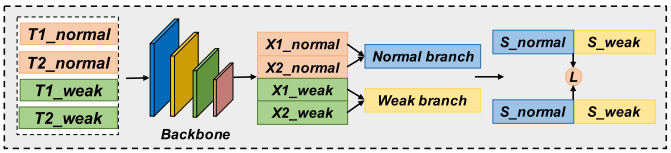


Fig. 5. Introduction of details of the proposed framework training and inferencing, where L represents Loss.

dataset k, and $y_{i,j}^{n^{(m)}}$ and $y_{i,j}^{w^{(m)}}$ represent the label of each pixel of the input image. The total loss L_{total} can be expressed as

$$L_{\text{total}} = L_{D_n} + L_{D_w}. \tag{8}$$

F. Training and Inference Details

In this section, we provide training details and the proposed framework's inferencing details.

Prior to training, strongly labeled pixel-level samples are resampled to balance the number of strongly and weakly labeled samples in the dataset. For instance, if strongly labeled samples represent 1/128 of the dataset and weakly labeled samples represent 127/128, the strongly labeled samples are repeated 127 times to achieve balance.

As shown in Fig. 5, During training, both strongly labeled data ($T1_{\text{normal}}$ and $T2_{\text{normal}}$) and weakly labeled data ($T1_{\text{weak}}$ and $T2_{\text{weak}}$) are simultaneously input into the feature extraction network to simplify the training process. The extracted features are fed into the normal and weak branches to produce separate network outputs (S_{normal} and S_{weak}), and the network is updated by computing separate loss functions for the strong and weak labels. During inference, the optimized feature extraction network only uses the normal branch.

IV. Experiments

In this section, we present the experimental setup for evaluating the proposed change detection algorithm. We begin by describing the two datasets used for the evaluation, namely WHU [71] and SYSU-CD [72], followed by the introduction of seven state-of-the-art fully supervised and semi-supervised change detection methods. Next, we provide implementation details for the training process and subsequently analyze the results of our algorithm on the two datasets both quantitatively and qualitatively. Finally, we discuss the effectiveness of our proposed methods.

A. Dataset Description

1) WHU [71]: This dataset consists of aerial images captured in April 2012, covering an area of 20.5 km^2 and containing 12 796 buildings (16 077 buildings in the 2016 version). Some sample images are presented in fig. 6(a). The subdataset was created by manually selecting 30 ground control points to geo-rectify the original dataset with an accuracy of 1.6 pixels. The subdataset and the corresponding images, along with building vector and raster maps, are now publicly available. The high-resolution aerial image has a size of 32 507 × 15 354. No specific data decomposition scheme is provided in [71], but we partitioned the image into nonoverlapping 512 × 512 pixel blocks using a sliding window and split them into training, validation, and test sets, which contain 1189, 319, and 319 image block pairs, respectively.

2) SYSU-CD [72]: The dataset contains 20 000 pairs of 0.5-m resolution aerial images captured in Hong Kong between 2007 and 2014. Sample images are shown in Fig. 6(b). In [72], the 800 original image pairs of size 1024 × 1024 were divided into training, validation, and test sets in a 6:2:2 ratio. Then, 25 image subset pairs of size 256 × 256 were randomly selected from each image pair and augmented by random flips and rotations to obtain a total of 20 000 pairs of aerial image patches. The dataset includes various types of significant changes, such as new urban construction, suburban sprawl, preconstruction foundations, vegetation changes, road widening, and marine construction.

B. Comparison Method

1) FC-Siam-Diff [42]: A feature fusion method that uses a siamese full CNN to extract multilevel features and use feature differences to fuse bitemporal information.

2) DTCDSCN [45]: A multiscale feature fusion method combines the channel attention mechanism and spatial attention mechanism in FCN to obtain more discriminative features.

3) SNUNet [46]: A densely connected siamese NestedUnet for change detection. It mitigates the local information loss in the deeper layers of the neural network through compact information transmission from the encoder to the decoder.

4) DSIFN [47]: A deeply supervised image fusion network that first uses a two-branch FCN for feature extraction and then uses a deeply supervised difference discriminative network (DNN) for change detection. To strengthen the completeness of change map boundaries and internal density, multilevel semantic features are blended with image differential map features by means of an attention mechanism.

5) BIT_CD [52]: A change detection method based on Transformers. It uses CNN to extract deep features and feed them into two branch transformers. Then the original bitemporal features are enhanced and the pixelwise absolute of the difference between the two branch feature maps is generated. Finally, several convolutions are leveraged to get the change map.

6) s4GAN [54]: An adversarial semi-supervised semantic segmentation method. It consists of a segmentor and a discriminator, the segmentor and the discriminator are trained in an adversarial way. Two proposed losses stabilize low-data training and balance the segmentor and discriminator.

7) SemiCD [57]: A semi-supervised method for change detection. It contains a fully supervised training phase and an unsupervised training phase. After the normal fully supervised training phase, in the unsupervised stage, the network constrains the output change maps to be consistent under different perturbations.

C. Evaluation Metrics

The F1 is the weighted reconciled average of precision and recall, which considers both precision and recall to balance the conflict and can better reflect the change detection ability of the model. So we use the average F1 score of change

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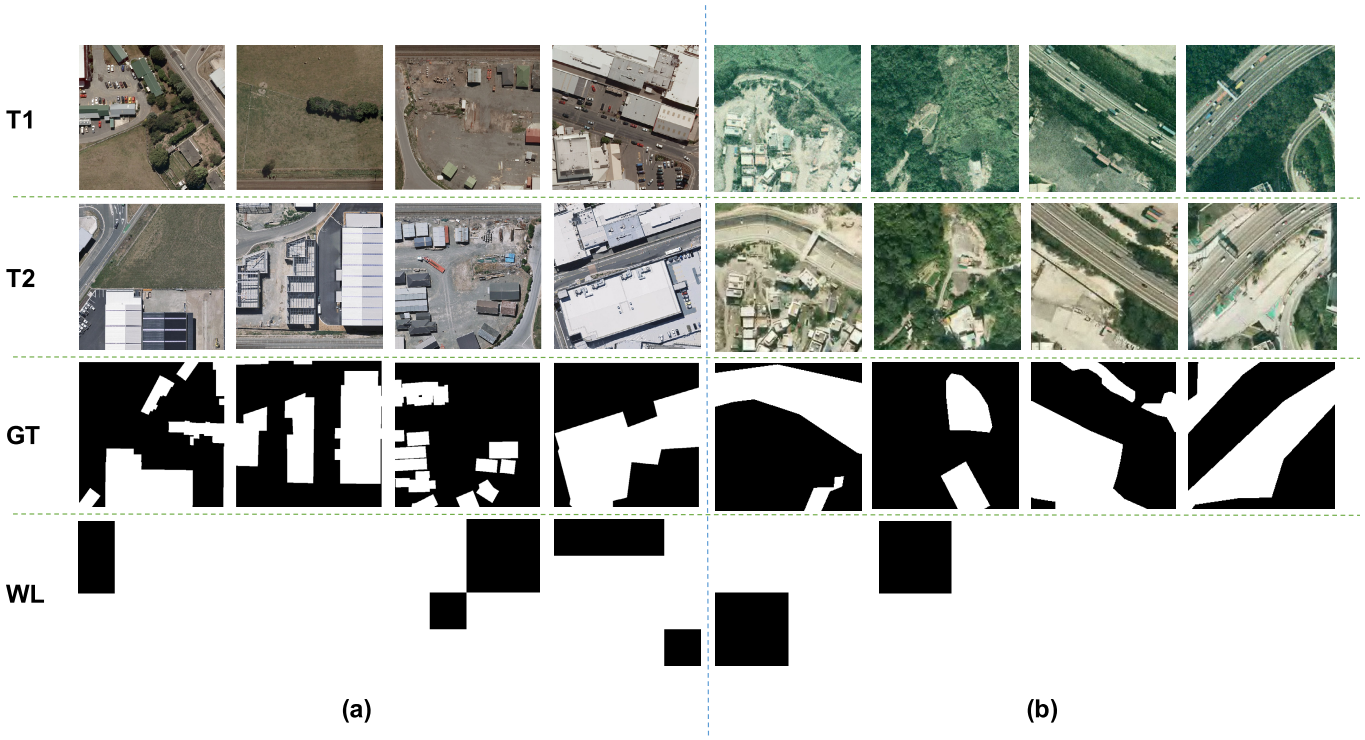


Fig. 6. Sample diachronic images and ground truth from (a) WHU Dataset and (b) SYSU-CD change detection datasets. The first row indicates the image of the T1 time phase, the second row shows the image of the T2 time phase, and the third row means the ground truth. The last line is the weak label generated by the SWL method proposed in this article.

category and background as the main evaluation index, which is calculated publicly as follows:

$$\mathrm{F1} = \frac{2 \times \mathrm{precision} \times \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}. \tag{9}$$

In addition, we also used Precision, Recall, and Intersection over Union (IoU) as auxiliary evaluation metrics, which are calculated as follows:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{10}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{11}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}}. \tag{12}$$

Among them, TP, FP, TN, and FN are the number of true positives, false positives, true negatives, and false negatives.

D. Experiment Setup

To train the SDCDNet proposed in this article, we utilized the BCE loss function [70] and stochastic gradient descent (SGD) optimizer [73] with momentum set to 0.9. We initialized the learning rate to 0.01 and employed a warm-up strategy [61] to gradually increase it to the preset value during the first 5 epochs. Subsequently, we applied cosine annealing to decay the learning rate and set the weight decay to 0.0005. The models were trained for 100 epochs with a minibatch size of 8, using the PyTorch [74] deep learning framework across four Nvidia 2080Ti GPUs.

TABLE I

RESULTS OF ABLATION EXPERIMENTS OF THE PROPOSED FRAMEWORK ON THE SYSU-CD DATASET, WITH THE HIGHEST SCORES MARKED IN BOLD AND ALL SCORES DESCRIBED IN PERCENTAGES

Method	Precision	Recall	F1 score	MIoU	Parameters(M)
Baseline	84.37	86.63	86.50	76.89	72
Baseline+AWM	84.92	87.29	87.05	77.33	72+3.5
Baseline+MCM	85.07	87.19	87.11	77.49	72+2.8
Baseline+AWM+MCM	**85.64**	**87.97**	**87.52**	**78.26**	72+6.3

E. Ablation Studies

To investigate the impact of the AWM and MCM on remote sensing image change detection with our proposed SDCDNet, we used DSIFN as the base model to create a baseline for ablation experiments on the SYSU-CD dataset [72]. We used a patch size of 128 pixels and pixel-level strong annotations at a scale of 1/8. This section presents the results of our ablation experiments, which were quantitatively evaluated using five comprehensive metrics: Precision, Recall, F1 score, mean Intersection over Union (MIoU), and Parameters.

Our experiments confirmed the usefulness of both the AWM and MCM. As shown in Table I, the addition of the AWM improved all four evaluation metrics, demonstrating that the adaptive fusion of features at all levels enhanced focus on small change regions. On the other hand, the MCM, which prioritizes foreground information, improved Precision, F1 score, and MIoU compared to the AWM, while slightly reducing Recall due to the negative impact of too few samples on change buildings. When both modules were added together,

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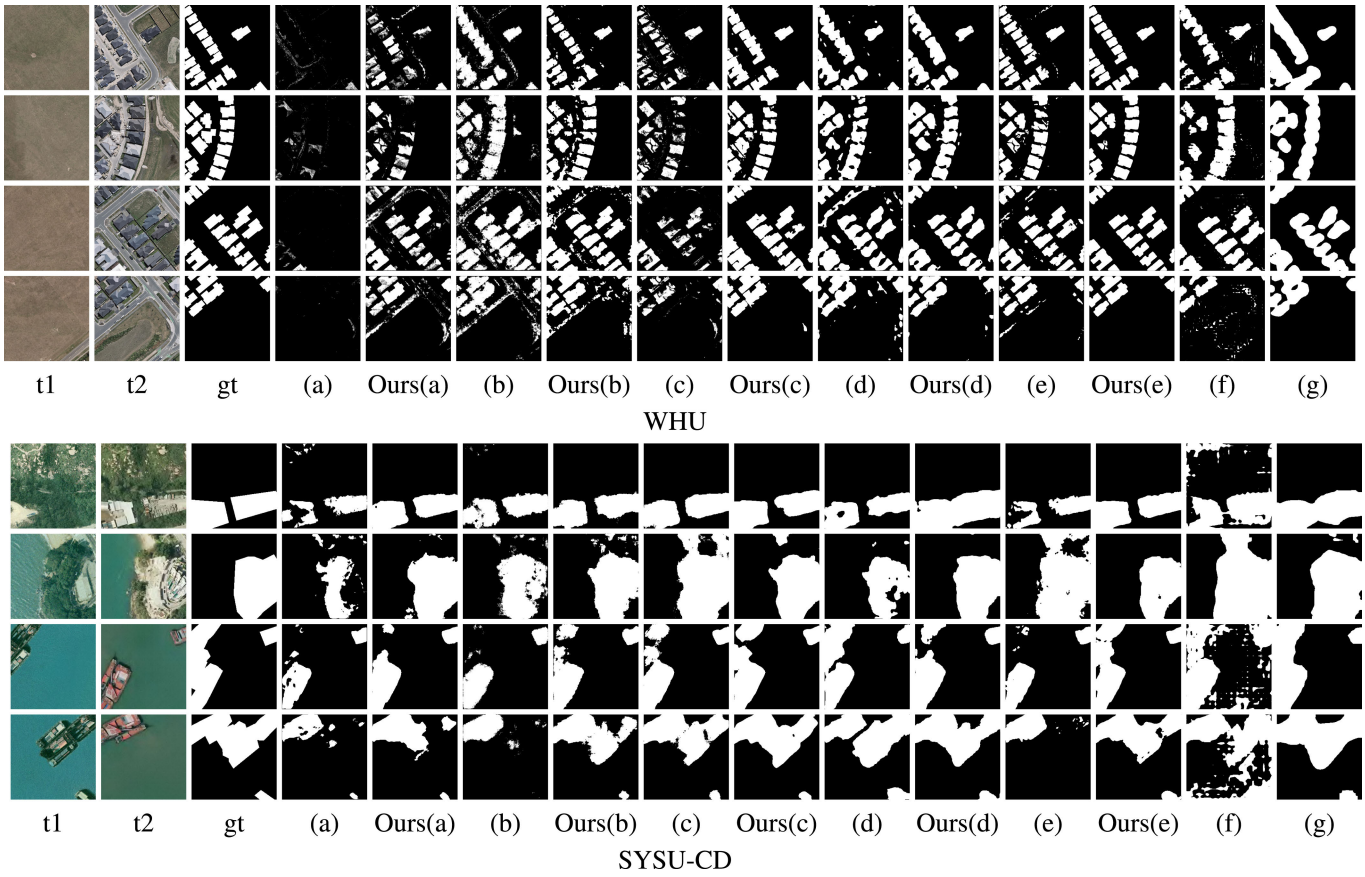


Fig. 7. Qualitative comparison results of change detection in the WHU and SYSU-CD datasets. (t1) t1 images. (t2) t2 images. (gt) Ground truth. (a) FC-Siam-Diff. (b) DTCDSCN. (c) SNUNet. (d) BIT_CD. (e) DSIFN. (f) s4GAN. (g) SemiCD. The changed area is white and the unchanged area is black.

the model's learning ability was strengthened, resulting in 1.51% improvement in Precision, 1.55% improvement in Recall, 1.18% improvement in F1 score, and 1.78% improvement in MIoU compared to the baseline. The additional parameters required by the AWM and MCM were 3.5M and 2.8M, respectively, compared to the baseline. These ablation experiments demonstrate the effectiveness of the proposed AWM and MCM, and the increase in computational cost is acceptable. The modules enhance the network's ability to extract foreground information features and perform multi-scale fusion, resulting in more complete detection of change boundaries and an overall improvement of the network's shortcomings.

F. Experimental Comparison and Analysis

To evaluate the superiority of the proposed framework SDCDNet, four state-of-the-art fully supervised change detection models were used as base models on WHU [71] and SYSU-CD [72] datasets. The performance of all methods was quantitatively and qualitatively compared with the four base models, including FC-Siam-Diff [42], DTCDSCN [45], SNUNet [46], BIT_CD [52], and DSIFN [47], using the proposed framework. In addition, to provide a comprehensive analysis, we also compared our method with state-of-the-art semi-supervised change detection methods and segmentation methods, namely SemiCD [57] and s4GAN [54] (s4GAN was implemented in the context of change detection).

1) Qualitative Comparison: To compare the visual performance of the different methods, Fig. 7 presents a qualitative comparison of our framework with other state-of-the-art fully supervised base models on the WHU and SYSU-CD datasets. As shown in Fig. 7, our proposed method outperforms the base models in detail processing for small target detection, significantly reducing missed and erroneous detections. For large targets, our method shows better detection continuity compared to the base models. Thus, our proposed semi-supervised SDCDNet framework surpasses the base model approach in terms of visual effects.

2) Quantitative Comparison: To quantify the results, according to different SSL settings, we can conclude that the proposed framework achieves the best accuracy of this article in all SSL settings compared to the base model under the condition that the strongly labeled data is only 1/8. As shown in Tables II and III, the improvement is especially significant when the Patch Size becomes smaller. For instance, on the WHU Building Dataset, when the Patch Size is 128 pixels, our proposed framework outperforms the base model DSIFN by 9.64%, 7.45%, 7.44%, and 11.29% in Precision, Recall, F1, and MIoU, respectively. Similarly, on the SYSU-CD Dataset, our framework shows improvements of 4.3%, 4.86%, 6.12%, and 8.85% in Precision, Recall, F1, and MIoU, respectively, compared to the base model DSIFN. The performance further improves when the Patch Size is reduced to 16 pixels. In the WHU Building Dataset, our proposed

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TABLE II

QUANTITATIVE COMPARISON OF CHANGE DETECTION METHODS WITH DIFFERENT LEVELS OF PATCH SIZE ON THE WHU BUILDING DATASETS, THE LABELING RATIO SET TO 1/8, ALL SCORES DESCRIBED IN PERCENTAGES

Patch Size	Method	Precision	Recall	F1	MIoU
——	FC-Siam-Diff	72.05	67.58	69.52	60.68
	DTCDSCN	82.66	66.14	71.24	61.70
	SNUNet	78.00	76.62	77.75	68.46
	BIT_CD	92.90	73.70	80.13	70.62
	DSIFN	82.68	88.95	86.52	78.18
16px	Ours(FC-Siam-Diff)	80.37	79.00	80.51	70.32
	Ours(DTCDSCN)	88.43	75.68	80.62	71.26
	Ours(SNUNet)	91.24	89.07	89.12	82.29
	Ours(BIT_CD)	94.79	86.05	89.90	82.96
	Ours(DSIFN)	92.55	97.51	94.82	89.99
32px	Ours(FC-Siam-Diff)	79.84	78.27	79.77	69.80
	Ours(DTCDSCN)	88.40	75.18	80.52	71.07
	Ours(SNUNet)	89.93	88.48	88.24	80.98
	Ours(BIT_CD)	91.08	88.26	88.61	82.57
	Ours(DSIFN)	91.49	96.78	93.86	88.63
64px	Ours(FC-Siam-Diff)	79.26	77.53	78.69	69.18
	Ours(DTCDSCN)	88.12	74.31	79.65	69.84
	Ours(SNUNet)	86.66	87.61	87.27	79.81
	Ours(BIT_CD)	92.06	82.69	86.72	78.62
	Ours(DSIFN)	90.88	96.05	93.28	87.67
128px	Ours(FC-Siam-Diff)	78.89	77.45	78.15	68.79
	Ours(DTCDSCN)	88.04	73.44	78.81	68.31
	Ours(SNUNet)	85.48	87.49	86.45	78.07
	Ours(BIT_CD)	89.67	81.45	85.03	76.49
	Ours(DSIFN)	90.65	95.58	92.96	87.01

TABLE III

QUANTITATIVE COMPARISON OF CHANGE DETECTION METHODS WITH DIFFERENT LEVELS OF PATCH SIZE ON THE SYSU-CD DATASETS, THE LABELING RATIO SET TO 1/8, ALL SCORES DESCRIBED IN PERCENTAGES

Patch Size	Method	Precision	Recall	F1	MIoU
——	FC-Siam-Diff	67.21	77.49	70.55	60.00
	DTCDSCN	75.29	80.40	75.59	64.30
	SNUNet	79.81	80.78	79.28	69.76
	BIT_CD	79.20	83.25	80.14	69.51
	DSIFN	82.11	83.89	82.47	71.90
16px	Ours(FC-Siam-Diff)	78.82	89.64	81.54	69.73
	Ours(DTCDSCN)	80.10	84.55	81.72	70.22
	Ours(SNUNet)	85.50	87.11	86.06	76.23
	Ours(BIT_CD)	85.48	84.88	85.18	74.98
	Ours(DSIFN)	86.33	88.71	88.12	78.87
32px	Ours(FC-Siam-Diff)	78.41	89.09	80.75	69.25
	Ours(DTCDSCN)	79.15	84.47	81.20	70.08
	Ours(SNUNet)	84.99	86.68	85.92	75.60
	Ours(BIT_CD)	85.18	84.37	94.76	74.38
	Ours(DSIFN)	86.01	88.51	87.88	78.49
64px	Ours(FC-Siam-Diff)	78.02	88.99	80.61	69.22
	Ours(DTCDSCN)	78.63	84.09	80.79	69.36
	Ours(SNUNet)	84.71	86.39	85.59	75.44
	Ours(BIT_CD)	83.80	84.57	84.17	73.60
	Ours(DSIFN)	85.81	88.19	87.66	78.28
128px	Ours(FC-Siam-Diff)	77.50	88.46	80.38	68.69
	Ours(DTCDSCN)	78.33	83.57	80.43	68.73
	Ours(SNUNet)	84.49	86.23	85.31	75.23
	Ours(BIT_CD)	83.82	81.43	83.86	72.62
	Ours(DSIFN)	85.64	87.97	87.52	78.26

framework achieves improvements of 11.94%, 9.62%, 9.59%, and 15.11% in Precision, Recall, F1, and MIoU, respectively. In the SYSU-CD Dataset, our method shows improvements of 5.14%, 5.75%, 6.85%, and 9.69% in Precision, Recall, F1, and MIoU, respectively. This is because reducing the Patch Size leads to more accurate weak labels, which in turn helps the model achieve better accuracy. Moreover, even when using only 1/4 of the finely labeled samples, our method achieves an accuracy that is not lower than or even higher than the base model using all the data.

SemiSANet is a simple yet effective semi-supervised method for change detection that employs consistency regularization and strong augmentation [59]. In the experimental part of this article, it was found that on the WHU Building dataset, SemiSANet achieved F1 scores of 0.7808, 0.7944, 0.8353, and 0.8786, with labeled ratios of 1/40, 1/20, 1/10, and

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TABLE IV

QUANTITATIVE COMPARISON OF CHANGE DETECTION METHODS WITH DIFFERENT LEVELS OF LABELED RATIOS ON THE WHU BUILDING DATASETS, THE PATCH SIZE SET TO 128 PIXELS, WITH THE HIGHEST SCORES MARKED IN BOLD AND ALL SCORES DESCRIBED IN PERCENTAGES

Method	Labeled Ratio													
	1/1		1/4		1/8		1/16		1/32		1/64		1/128	
	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU
FC-Siam-Diff	78.47	69.06	73.49	65.16	69.52	60.68	65.81	58.42	59.69	53.02	54.69	50.02	51.31	48.25
Ours(FC-Siam-Diff)	—	—	**78.90**	**69.52**	**78.15**	**68.79**	**77.00**	**67.50**	**76.14**	**64.27**	**75.23**	**63.89**	**71.94**	**63.22**
DTCDSCN	77.18	67.54	71.90	62.21	71.24	61.70	70.23	60.6	68.52	58.29	63.25	54.07	60.21	52.43
Ours(DTCDSCN)	—	—	**79.26**	**69.49**	**78.81**	**68.31**	**78.22**	**67.62**	**77.33**	**67.71**	**73.09**	**63.50**	**70.07**	**61.46**
SNUNet	88.44	80.95	82.00	72.95	77.75	68.46	74.87	65.47	69.75	60.57	65.79	58.60	56.05	52.00
Ours(SNUNet)	—	—	**88.19**	**80.02**	**86.45**	**78.07**	**82.90**	**73.59**	**80.11**	**70.85**	**75.99**	**66.69**	**74.29**	**65.28**
BIT_CD	91.58	85.43	83.19	74.19	80.13	70.62	78.51	68.80	—	—	—	—	—	—
Ours(BIT_CD)	—	—	**88.04**	**81.26**	**85.03**	**76.49**	**84.89**	**76.32**	**77.52**	**67.64**	**73.63**	**63.59**	**69.22**	**59.26**
DSIFN	94.42	89.89	90.69	84.60	86.52	78.18	83.08	74.61	80.85	71.74	60.96	55.43	51.69	50.02
Ours(DSIFN)	—	—	**94.55**	**89.89**	**92.96**	**87.01**	**88.44**	**81.62**	**87.20**	**78.79**	**82.58**	**74.43**	**77.15**	**70.24**
s4GAN	—	—	80.61	71.88	78.35	64.83	65.46	60.26	63.44	59.12	63.15	58.46	59.81	58.20
SemiCD	—	—	86.68	75.10	79.56	70.17	78.05	68.65	77.21	67.84	74.77	65.56	57.09	52.27

TABLE V

QUANTITATIVE COMPARISON OF CHANGE DETECTION METHODS WITH DIFFERENT LEVELS OF LABELED RATIOS ON THE SYSU-CD DATASETS, THE PATCH SIZE SET TO 128 PIXELS, WITH THE HIGHEST SCORES MARKED IN BOLD AND ALL SCORES DESCRIBED IN PERCENTAGES

Method	Labeled Ratio													
	1/1		1/4		1/8		1/16		1/32		1/64		1/128	
	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU
FC-Siam-Diff	80.34	68.65	72.55	62.02	70.55	60.00	69.11	56.16	68.26	55.45	66.63	53.02	66.07	52.42
Ours(FC-Siam-Diff)	—	—	**81.90**	**70.52**	**80.38**	**68.69**	**79.80**	**67.92**	**79.57**	**67.59**	**79.06**	**66.85**	**78.15**	**66.03**
DTCDSCN	80.13	68.39	78.81	66.88	75.59	64.30	74.09	63.53	70.74	59.93	65.42	52.69	57.73	48.83
Ours(DTCDSCN)	—	—	**83.08**	**72.09**	**80.43**	**68.73**	**80.06**	**68.15**	**75.88**	**63.34**	**74.16**	**61.61**	**71.81**	**58.92**
SNUNet	84.51	74.08	80.63	70.93	79.28	69.76	75.99	65.99	71.99	63.07	70.28	60.65	65.00	55.97
Ours(SNUNet)	—	—	**85.51**	**75.45**	**85.31**	**75.23**	**83.29**	**72.88**	**81.59**	**70.08**	**79.49**	**67.55**	**77.31**	**64.82**
BIT_CD	85.53	75.52	82.90	72.60	80.14	69.51	78.44	66.52	76.26	63.34	75.73	62.66	70.95	57.24
Ours(BIT_CD)	—	—	**85.14**	**74.26**	**83.86**	**72.62**	**83.15**	**72.25**	**82.96**	**71.84**	**81.86**	**70.50**	**79.38**	**67.62**
DSIFN	87.96	79.12	84.76	75.29	82.47	71.90	79.79	67.83	76.17	62.59	72.37	60.0	68.44	55.91
Ours(DSIFN)	—	—	**87.88**	**78.78**	**87.52**	**78.26**	**86.38**	**76.79**	**85.62**	**75.60**	**83.88**	**73.18**	**80.70**	**69.04**
s4GAN	—	—	79.81	67.54	78.15	66.73	75.29	63.53	73.51	61.34	62.37	54.25	58.89	49.36
SemiCD	—	—	83.23	73.99	80.49	71.50	79.78	70.06	77.58	67.34	75.45	64.92	71.00	60.23

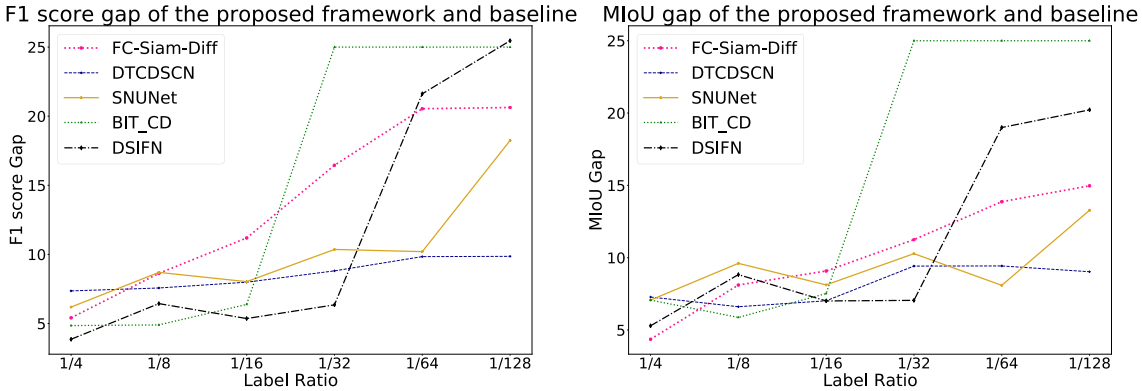


Fig. 8. Trend of the variance between the proposed framework and the base model in terms of F1 and MIoU scores as the labeling rate decreases on the WHU Building Dataset.

1/5, respectively. We propose a new method called SDCDNet, which builds on DSIFN. In our experiments, when the labeled ratio was divided into 1/64, 1/32, 1/16, and 1/8, SDCDNet outperformed SemiSANet in terms of F1 scores.

To demonstrate the superiority of our proposed framework, we compared it with the base model on two datasets, WHU Building Dataset and SYSU-CD Dataset, using a patch size of 128 pixels. The results, presented in Tables IV and V, indicate that our proposed method outperforms the base model for all labeling ratios. Additionally, the improvement of our proposed framework over the base model is more significant when the labeling rate is lower, as illustrated in Figs. 8 and 9. Specifically, on the WHU Building Dataset, our method achieved a 4.26% increase in F1 score and a 6.25% increase in MIoU over the base model DSIFN when the strongly labeled data is reduced to 1/4. On the SYSU-CD Dataset, our method achieved a 3.68% increase in F1 score and a 4.64% increase in MIoU. Notably, when trained with only 1/128 strongly

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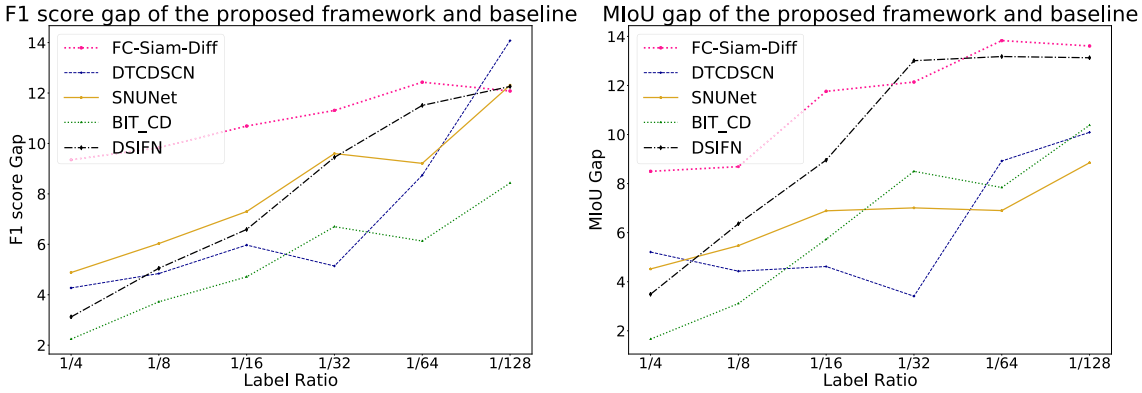


Fig. 9. Trend of the variance between the proposed framework and the base model in terms of F1 and MIoU scores as the labeling rate decreases on the SYSU-CD Dataset.

TABLE VI

COMPARISON STUDY OF MODEL EFFICIENCY. WE REPORT THE NUMBER OF PARAMETERS (PARAMETERS) AND FLOATING POINT OPERATIONS PER SECOND (FLOPS) FOR TWO SEMI-SUPERVISED MODELS AND THE PROPOSED SDCDNET MODEL ON FIVE BASE MODELS. THE INPUT OF THE MODEL IS ADJUSTED TO $256 \times 256 \times 3$ TO COMPUTE FLOATING POINT OPERATIONS

Method	Parameters (M)	FLOPs (G)
Ours(FC-Siam-Diff)	2.22	17.66
Ours(DTCDSCN)	41.07	50.22
Ours(SNUNet)	19.35	115.81
Ours(BIT_CD)	3.04	18.85
Ours(DSIFN)	71.45	171.32
s4GAN	139.86	183.83
SemiCD	46.85	146.17

TABLE VII

QUANTITATIVE COMPARISON OF CHANGE DETECTION METHODS WITH DIFFERENT LEVELS OF LABELED RATIOS ON THE GOOGLE DATASET, THE PATCH SIZE SET TO 128 PIXELS, WITH THE HIGHEST SCORES MARKED IN BOLD AND ALL SCORES DESCRIBED IN PERCENTAGES

Method	Labeled Ratio							
	1/1		1/8		1/32		1/128	
	F1	MIoU	F1	MIoU	F1	MIoU	F1	MIoU
DSIFN	94.99	90.84	82.73	72.73	74.65	64.04	69.31	58.91
Ours(DSIFN)	—	—	**90.85**	**85.78**	**83.39**	**73.16**	**79.29**	**68.76**
s4GAN	—	—	83.54	73.77	77.77	66.89	55.55	44.17
SemiCD	—	—	84.81	75.31	66.50	56.29	60.64	46.65

labeled data, our method achieved significant improvements in F1 score and MIoU on both datasets, with 34.28% and 40.42% increases on the WHU Building Dataset and 17.91% and 23.48% increases on the SYSU-CD Dataset, respectively. Our method also outperforms other semi-supervised models s4GAN and SemiCD on both datasets, especially when the proportion of pixel-level labeled samples is low.

Table VI presents a comparison of the number of parameters and computational effort for each model under the same condition. Our proposed method achieved superior performance compared to the semi-supervised methods on most base models, demonstrating its effectiveness.

We attribute this significant improvement to two main reasons. First, as the amount of strongly labeled data decreases, the base model's feature distribution will have a relatively large error with the real distribution. Our proposed framework resamples a small amount of strongly labeled data to allow the Normal branch to exploit the feature distribution information, and the MCM used in the Weak branch enables the model to extract compelling discriminative features from the weakly labeled data. Second, the consistency of the feature distribution between the Normal and Weak branches is continuously improved during training, leading to high accuracy with minimal strongly labeled data and a large amount of weakly labeled data.

3) Quantitative Analysis in Google Dataset: In order to further validate the performance of the proposed method on different datasets, we compared it with three representative methods on the Google dataset [75], where the changed objects vary in size and shape. As shown in Table VII, our proposed method exhibits significant advantages over other semi-supervised methods under different semi-supervised settings, demonstrating its high effectiveness.

4) Discussion and Analysis: Our proposed semi-supervised change detection method can effectively detect changes with limited labeling. However, there are two main limitations to our work. First, our method depends heavily on the performance of the fully supervised change detection method it is used with, which may affect the quality of the change detection results. Second, as discussed in Section III-F, when the labeling ratio is low, the number of strongly labeled samples must be repeatedly sampled to match the number of weakly labeled samples for alignment, which can increase the training time and impact the results differently depending on the labeling ratio.

V. CONCLUSION

Labeling remote sensing data is expensive, and obtaining these labels in real-world applications is both challenging and tedious. To address these issues, we did not focus on building a new, specific semi-supervised change detection network model. Instead, we proposed a new, general, and convenient change detection framework based on SSL. This framework requires only a small amount of strongly labeled data and a large amount of weakly labeled data to achieve highly accurate change detection, which becomes more apparent when the

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proportion of strongly labeled data in the training set is smaller. Additionally, if the existing fully supervised change detection model is used as the base model, the framework can quickly implement semi-supervised change detection training by adding Weak branches to the base model without changing its structure. The experimental results demonstrate the effectiveness of the proposed framework, with a significant improvement in performance observed when compared to the existing fully supervised change detection model under the same experimental data and conditions. In future research, we will explore the application of this framework on base models with stronger feature extraction abilities.

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